Pixel-Level Remote Sensing Image Recognition Based on Bidirectional Word Vectors

Hongfeng You, Shengwei Tian, Long Yu, and Yalong Lv

***Abstract*— In the traditional remote sensing image recognition, the traditional features (e.g., color features and texture features) cannot fully describe complex images, and the relationships between image pixels cannot be captured well. Using a single model or a traditional sequential joint model, it is easy to lose deep features during feature mining. This article proposes a new feature extraction method that uses the word embedding method from natural language processing to generate bidirectional real dense vectors to reflect the contextual relationships between the pixels. A bidirectional independent recurrent neural network (BiIndRNN) is combined with a convolutional neural network (CNN) to improve the sliced recurrent neural network (SRNN) algorithm model, which is then constructed in parallel with graph convolutional networks (GCNs) under an attention mechanism to fully exploit the deep features of images and to capture the semantic information of the context. This model is collectively named an improved SRNN and attention-treated GCN-based parallel (SAGP) model. Experiments conducted on Populus euphratica forests demonstrate that the proposed method outperforms traditional methods in terms of recognition accuracy. The validation done on public data set also proved it.**

***Index Terms*— Attention mechanism, bidirectional independent recurrent neural network (BiIndRNN), bidirectional word vector, graph convolutional networks (GCNs), parallel joint algorithm, sliced recurrent neural network (SRNN).**

I. Introduction

With the development of remote sensing technology, the spatial resolution of remote sensing images is increasing, and scholars are increasingly focusing on pixel-level detection in images [30]–[34]. With in-depth understanding of the pixel, researchers have found that the relationship between a pixel and the surrounding pixels plays a nonnegligible role [35]–[38] that can be used to improve the accuracy of image recognition.

In recent years, convolutional neural networks (CNNs) have played various irreplaceable roles in image recognition including sentiment analysis of faces [39], military detection [40], and medical image lesion detection [41]. CNNs [1], [2] can adaptively extract the most meaningful features of image pixels for identifying and classifying the target under detection. However, the graph convolutional network (GCN) [43]–[46] algorithm mainly considers and constructs the relationships between nodes (nodes can represent a word in a sentence, a pixel in an image, a feature in a pixel, and so on). In this article, the nodes represent the features of the image pixels. Thus, the use of nodes accounts for the lack of a CNN algorithm in this aspect.

Algorithms such as recurrent neural networks (RNNs) [3], [4] and long short-term memory (LSTM) [5], [6], which are commonly used in natural language processing, can handle contextual relationships well and store historical information. However, when a traditional RNN processes long sequence data, the front part of the features in the sequence are lost, resulting in an inability to completely capture the deep features [52], [53] and worsening the recognition effects. Multispectral images are very rich in features, and it is important for researchers to find a better algorithm to store the complete information of multispectral images.

Word embedding maps words into vectors and reflects the true relationships between words by calculating vectors. Research has shown that word vectors can better establish and preserve the relationships between words and help achieve satisfactory results in domains such as emotion analysis [47], text classification [48], performance analysis [49], and ontology enhancement [50].

Compared with the existing methods, the proposed model in this article can more effectively establish deep features and contextual semantic features around pixels by using word embedding techniques. The main contributions of this article can be summarized as follows.

1) Using the semantic relationships of natural language processing, a bidirectional word vector feature [8] that can more accurately describe the correlations between image pixels is proposed.
2) A bidirectional independent recurrent neural network (BiIndRNN) is proposed to improve the sliced recurrent neural network (SRNN) [9]. This network can store additional information and the contextual relationships between the features of image pixels. The last layer of the SRNN algorithm is replaced by a CNN to fully capture the original features of the image.
3) An attention [17], [18] mechanism is added to enhance the probability weight of the recognition target, and the

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GCN [20] algorithm is then used to identify the directional relationship between the targets while extracting the deep features to establish the positional relationships between the features of image pixels.

4) The improved SRNN algorithm and the attention-GCNs run in parallel with the algorithm [29], replacing the traditional sequential joint model algorithm; as a result, the image features and context relationships identified are more detailed and complete and are deeper.

The remainder of this article is organized as follows. In Section II, two directions for image recognition are discussed: feature engineering and deep learning algorithms. The proposed improved SRNN and attention-treated GCN-based parallel (SAGP) model and the related details are presented in Section III. The simulated and real data experimental results and a discussion are presented in Section IV. Finally, our conclusions are given in Section V.

II. Related Research

Currently, image classification is mainly based on deep learning and feature engineering. Because the features of remote sensing images are becoming more abundant, traditional image feature-based (e.g., color features and texture features) methods and even single deep learning algorithms have difficulty in extracting features that fully represent the entire image. Therefore, to fully capture the potential features of an image and the semantic relationships between the image pixels, researchers have conducted extensive study on feature engineering and deep learning algorithms.

Based on the research on feature engineering, the pixels, shapes, and regions in an image are important for recognition. Zhu *et al.* [10] effectively combined global features and texture features of remote sensing images to implement a global feature scene classifier that yields more accurate results, which thereby fully illustrated the importance of global features. Lord and Reed [11] demonstrated that information differs between pixels in an image, which are assigned a weight function associated with their distance to alleviate the impact from the surroundings pixels. Zanotta *et al.* [12] proposed a multisize segmentation method for chopping images into segments of various sizes, and the importance of the segmentation size in image classification is explained in terms of registration. The speeded up robust feature (SURF) algorithm was used to determine the main directions of the features and generate feature descriptors. In these experiments, images were represented and understood by the features of traditional rules, and the potential features of the images were further explored by establishing various correlations between the pixels. According to the above research, by establishing the correlations between the pixels and determining the direction of each pixel, the last extracted features can better meet the experimental needs.

Regarding the research on deep learning, Cheng *et al.* [14] demonstrated that CNNs exhibited strong recognition performance for remote sensing image classification; these researchers evaluated the feasibility of this method on three public data sets. Zhang *et al.* [15] translated the spatial transformation information of remote sensing images with visual information into a natural language that conforms to a specified grammar and sentence structure and then learned the abstract features of the images via bidirectional long short-term memory (BiLSTM) [21] networks. These researchers demonstrated the feasibility of the method using two public data sets. Mou and Zhu [16] and others combined a CNN and an RNN into a single end-to-end sequential network. The former generated rich spectral–spatial feature representations, and the latter effectively analyzed the temporal dependency in bitemporal images. The joint model was used to capture the global feature information and outperformed each individual model. Although the sequential algorithms were able to perform joint feature extraction, the latter step in the sequential algorithms performed a second extraction based on the feature that was extracted by the preceding algorithm, and therefore, this model could not extract all the deep features.

Although these approaches can describe and express the complex content of images, these models are insufficient for semantic mining and making full use of the features. In the learning process, certain deep features are often lost. Therefore, due to these shortcomings, the image features are expanded and coded via word embedding technology to generate real dense vectors in this article. An attention mechanism [42] is used to enhance the probability weights of a target, and GCNs are used to establish the directional relationship between the weights, while the improved SRNN jointly extracts, captures, and learns the image features and ultimately integrates with the contextual semantic information. The results demonstrate that the proposed method outperforms the traditional deep learning methods.

III. SAGP Model Based on Bidirectional Glove

A. Summary of This Section

Section III is divided into six parts. In Section III-A, we summarize Section III to aid reader understanding. Section III-B mainly introduces the SAGP model algorithm and the model learning process. In Section III-C, we introduce the new word vector feature that we extract, which is called the bidirectional word vector feature, and in Section III-D, we introduce certain existing algorithms and the advantages of these algorithms. In Section III-E, we introduce our improved algorithm and the main role of this algorithm in detail. In Section III-F, we merge the deep features of each module.

B. Framework of the Model

As shown in Fig. 1, the SAGP model consists of three stages. In stage 1, the hue, saturation, and value (HSV) features and texture features of the image are retrieved. The spatial relationship information is translated into a natural language word vector as represented by a bidirectional word vector. In stage 2, the mixed features are input into an improved SRNN algorithm module, where an improved IndRNN [51] denoted BiIndRNN is used to learn the long sequential text; subsequently, a CNN is used to learn the deep features of the original image. The mixed features are simultaneously used to

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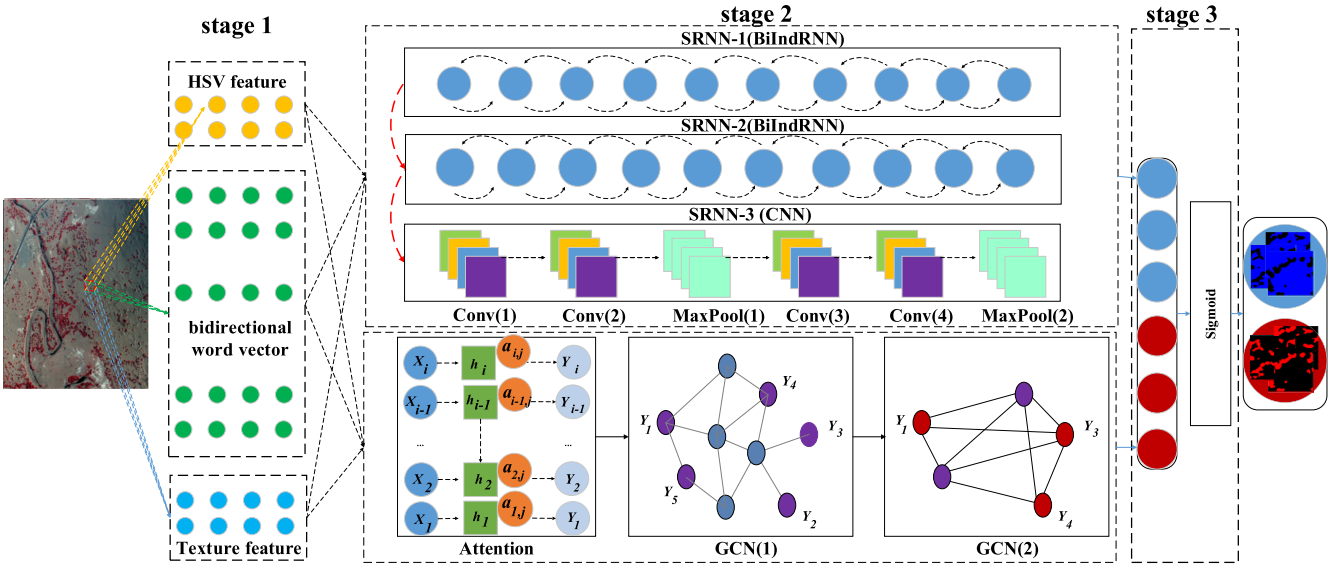


Fig. 1. Framework of the SAGP.

enhance the probability weight of the target on the image by using the attention mechanism to obtain the deep features. The GCN algorithm is then used to locate the positions of the deep features and establish the directions of the feature. The attention mechanism and the GCN algorithm constitute the attention-treated GCN module that runs in parallel with the SRNN module in stage 2, and the outputs of this module are merged in stage 3 to produce the final result.

C. Word Embedding

A word vector is a form that allows words in a natural language to be recognized and understood by computers. Barkan [19] proposed a word embedding by word vectors for mapping words to a low-dimensional space, which solved the vector sparsity problem. In the low-dimensional space, the mapping between words can reflect their semantic connection. To explore the relationship between the pixels in complex remote sensing images, this article proposes a bidirectional word embedding method. In this method, the Skip-gram [22], [23] algorithm in the Word2Vec and Glove [7] word embedding algorithms was used to generate real-time vector sets according to the horizontal and vertical directions of the spatial relationship features of an image, fully considering the current pixel and the five surrounding pixels on the left, right, above, and below to establish a relationship (see Fig. 2). In Fig. 2, the black circle marked with "1" represents the current pixel, the green circles marked with "2" represent the horizontal real vector, and the orange circles marked with "2" represent the vertical real vector; all the "2"s in both colors represent the surrounding pixels of the current pixel.

Skip-gram is a local context window method. If global statistics on the pixels is not available, this method assigns too much weight to high-exposure pixel values, while Glove considers the transcendental statistics of global pixels and uses a weight equation, namely, $f(X_{i,j})$, to control the relative weights of the pixels. The importance of global relationships is fully explained by comparing the two algorithms. The Glove

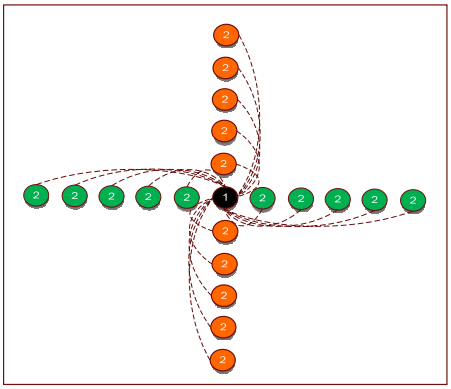


Fig. 2. Bidirectional Glove.

loss function is expressed as follows:

$$J = \sum_{i,j=1}^{V} f(X_{i,j})\left(w_i^T \tilde{w}_j + b_i + \tilde{b}_j - \log(X_{ij})\right)^2 \quad (1)$$

where $X_{i,j}$ is the number of times that pixel j appears in the window of pixel i in the two directions of the pixel, w_i^T represents the word vector of pixel i (corresponding to the black circle marked with "1"), $\tilde{w}_j$ represents the context word vector of pixel i, j represents the subscript of any pixel in the context window of the matrix (except i, equivalent to the circles marked with "2"), b_i is the scalar deviation of w_i^T, $\tilde{b}_j$ is the scalar deviation of $\tilde{w}_j$, and V is the total number of pixels in the dictionary.

D. Related Existing Algorithms or Methods

1) Independently Recurrent Neural Network: Although the RNN can process all the time-series information, it is difficult to completely record the context features in long-text data. The longer the learning text is, the more difficult it is to obtain the feature information from the preceding learning step, and the final model can only learn the information of subsequent content in the text; moreover, gradient disappearance and

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gradient explosion may occur in the process of multilayer training. As an improved RNN model, LSTM and gated recurrent unit (GRU) [24], [25] overcome the vanishing gradient and exploding gradient. However, as the number of layers in the stacked model increases, the hyperbolic tangent and sigmoid functions are used in LSTM, and GRU causes gradient attenuation. Thus, we used the IndRNN algorithm, which overcomes the vanishing gradient and exploding gradient. This model is effective for natural language text that is generated via computer vision. The IndRNN equation is illustrated as follows:

$$h_t = \sigma(\omega x_t + \lambda \cdot h_{t-1} + b) \tag{2}$$

where λ represents the cyclic weight vector, $\cdot$ represents the Hadamard product, ω and b represent the weight and bias terms, respectively, and h_{t-1} represents the hidden-layer state at time $t-1$. The independent RNNs use nonsaturation functions such as rectified linear units (ReLUs) to enhance the robustness. The neurons in each layer of IndRNN are independent, and the subsequent layer of neurons only accepts the information of the upper input and its current hidden state. The unique way of IndRNN processing neurons matches the direction of the label that is generated by the image pixels of the GCN model, which fully satisfies the experimental requirements. Equation (2) can be generally represented as follows:

$$h_{t,n} = \sigma(\omega_n x_t + \lambda_n \cdot h_{t-1,n} + b_n) \tag{3}$$

where ω_n, λ_n represent the current weights of the nth line and $h_{t,n}$ represents the nth hidden-layer state at time t.

2) Attention Mechanism: An attention mechanism is a type of probability weight distribution mechanism. By calculating the features of remote sensing image input at different times, this mechanism pays more attention to the features of target recognition. Thus, these features are assigned larger probability weights, which contribute to improving the quality of the hidden-layer features. The traditional features and the bidirectional word embedded features are used as the initial input of the attention-based mechanism to enhance the identification of related attributes, as shown in Fig. 1. The weight coefficient of the hidden layer is calculated as follows:

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{n} \exp(e_{ik})} \tag{4}$$

where a_{ij} represents the attention allocation coefficient of the feature sequence of the pixel j in the source input, i represents the moment, and e represents the energy value of the ith moment

$$c_i = \sum_{j=1}^{n} a_{ij} h_j. \tag{5}$$

h_j represents the hidden-state information of the pixel j by feature vectors. Finally, we use summation functions for a_{ij} and the hidden state h_j to generate the context vector c_i.

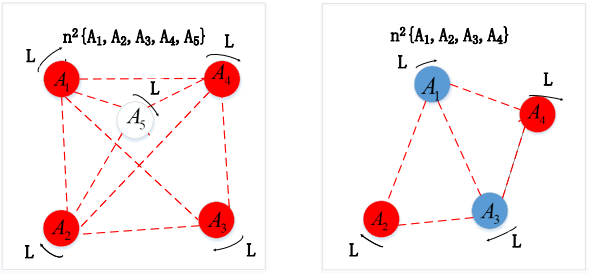


Fig. 3. Illustration of GCN feature selection.

3) Graph Convolutional Network: The feature that has the highest probability weight in the output of the attention mechanism is the input to the GCN, where the positional and directional correlations between deep features are established to generate direction labels. Via graph convolution, the most critical units are used to further extract the deep features, and the equation that is used in the first layer of the graph is expressed as follows:

$$h_v = \text{ReLU}\left(\sum_{u \in N(v)} \omega X_u + b\right) \tag{6}$$

where X_u is the set of the most critical units for storing the features with the highest weight. $N_{(v)}$ is the set of attention features, and h_v is the output of the current layer and the convolution input of the subsequent layer.

The k-layer is provided, and the equation for the kth layer convolution is represented as follows:

$$h_v^{k+1} = \text{ReLU}\left(\sum_{u \in N(v)} \omega^{(k)} h_u^{(k)} + b^{(k)}\right). \tag{7}$$

The traditional convolution algorithm is based on learning features that are in a fixed form, while the GCN algorithm generates the corresponding direction labels by inputting deep features and the positions of these features. Suppose there are multiple features with high-probability weights for the data. Let the pairwise features be denoted A1 and A2, then the tuples (A1, A2) and (A2, A1) belong to the direction information of this pixel. The GCN records the labels of the tuples as L(A1, A2) and L(A2, A1), respectively. The labels represent the context relationship between the features in pixel and indicate whether each edge is oriented to the left or right relative to the input. The feature position is illustrated as L in Fig. 3. Equation (7) is rewritten as (8). In this article, $k+1$ is equal to 2

$$h_v^{k+1} = \text{ReLU}\left(\sum_{u \in N(v)} \omega_L^{(k)} h_u^{(k)} + b_L^{(k)}\right). \tag{8}$$

As illustrated in Fig. 3, the execution order of the feature selection by multilayer GCNs is from the left subgraph to the right subgraph. The left subgraph represents the GCN model of the first layer, and the right subgraph represents the GCN model of the second layer. The red circles A1, A2, A3, and A4 and the colorless circle A5 in the left subgraph represent the feature points that have higher probability weights in the attention mechanism. However, the circle in red has a

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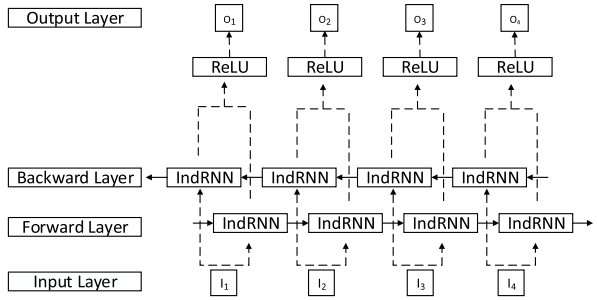


Fig. 4. BiIndRNN diagram.

higher probability weight than the colorless circle, and thus, the red circles are used as the input to the next layer of GCN (as shown in the right subgraph). The two blue circles in the right subgraph have lower weights than the red circles; hence, the two red circles are used as the feature input to the subsequent layer (the merge layer in the SAGP model) to mine deeper image features. Each of the dashed lines is bidirectional and represents the positional relationship between the features. In Fig. 3, n is a parameter that needs to be set at each layer of GCN to generate n^2 deep units.

4) Sliced Recurrent Neural Network: Traditional RNNs cannot simultaneously train multiple features. However, an SRNN implements parallel training to enable the exploration of more features. The principle of SRNNs is explained in Section III-E2.

E. Improvements in the SAGP Model

1) Bidirectional Independently Recurrent Neural Network: The LSTM sequence information refers only to a single propagation to the destination. The unidirectional mechanism only contains the forward information of the current feature. To overcome this problem, the BiLSTM was proposed to fully capture the information of the context features. Inspired by this idea, we propose a BiIndRNN that overcomes the vanishing gradient and exploding gradient and fully learns the information of the context feature.

The BiIndRNN equations are as follows:

$$\mathrm{Ind}_t = f(\omega_1 x_t + \omega_2 \mathrm{Ind}_{t-1}) \tag{9}$$

$$\mathrm{Ind}'_t = f(\omega_3 x_t + \omega_5 \mathrm{Ind}_{t+1}) \tag{10}$$

$$O_t = g(\omega_4 \mathrm{Ind}_t + \omega_6 \mathrm{Ind}'_t). \tag{11}$$

A diagram of the BiIndRNN is illustrated in Fig. 4.

As shown in Fig. 4, I_1, I_2, I_3, and I_4 in the input layer represent the mixed features of the input image. The forward layer is used to store the forward timing information, and the backward layer is used to store the reverse timing information. The robustness is enhanced by the nonlinear activation function ReLU. The output layer produces the output corresponding to the input.

2) Improved SRNN: We use the BiIndRNN described in Section III-E1 as the subsequence model of the SRNN model. The independent loop units in each layer simultaneously learn the timing information of the features, and the last layer changes to a CNN for mining the deep features from the original image. The output of the current layer is the input to the subsequent layer. The algorithm model formulas are as follows:

$$h_t^1 = \text{Bi-}\overrightarrow{\text{IndRNN}}^0\left(\text{mss}^0_{(t-l_0+1)\sim t}\right) \tag{12}$$

$$h_t^2 = \text{Bi-}\overrightarrow{\text{IndRNN}}^1\left(h^1_{t-l_1} \sim h^1_t\right) \tag{13}$$

$$C_t^3 = \overrightarrow{\text{CNN}}^2\left(h^2_{t-l_2} \sim h^2_t\right) \tag{14}$$

where mss represents the smallest sequence in the first layer, l_0 represents the length of the smallest sequence of the first layer, t represents the total length of multiple smallest sequences in the first layer, l_1 represents the smallest sequence length of h_t^1 by the BiIndRNN algorithm, and the result h_t^2 is obtained by the same principle as (12). The principle of the parameters in (14) is the same as that of the parameters in (12) and (13), but the method used to extract features is a CNN. In this article, a two-layer BiIndRNN is used to capture the relationship between the image mapping and the word vector, while the last layer of the CNN is used to mine the deep features of the image so that the real image vector and the real word vector are fully matched and ready for feature fusing.

F. Fusion of the Improved SRNN With the Attention-GCNs

The improved SRNN algorithm retains the semantic information features of the context and the original image features, which is denoted by C_t^3. GCN algorithm mines high-weight features and reserved the relationship between their features, which is denoted by h_v^2. We do feature splicing and fusion of C_t^3 and h_v^2 through the concatenate layer. Finally, the fusion features are classified by the sigmoid function. The fusion equation is expressed as follows:

$$\text{Merge}\left(\left[C_t^3, h_v^2\right]\right)). \tag{15}$$

IV. Data Set and Analysis of the Experimental Results

A. Data Set

The data used in this article were captured by the Quick Bird satellite (Fig. 5). One set of the experimental data regards the Populus euphratica forest on the northern margin of the Tarim Basin in Luntai County, Xinjiang, China, which is processed as data set 1. The images are sized 2000 × 1400 pixels. The Populus euphratica forest in the Tarim River Basin is processed as data set 2. The images are sized 2000 × 1400 pixels. This article studies a binary classification problem: the positive sample is Populus euphratica forest and the negative sample is the background (desert, tree shadow, highway, etc.), and the images were captured at the end of October 2005. Data set 1 and data set 2 both have 165 336 image blocks as samples. In the experiment, four-fifths of the sample is used as the training set, and one-fifth is used as the test set. All the remote sensing images are preprocessed, including fusion, atmospheric correction, geometric correction, and sharpening [26]–[28]. To further evaluate the generalization ability of the feature engineering approach and the SAGP model, several images from the public data set NWPU-RESISC45 are also used, as shown in Fig. 6. The experimental results and specific

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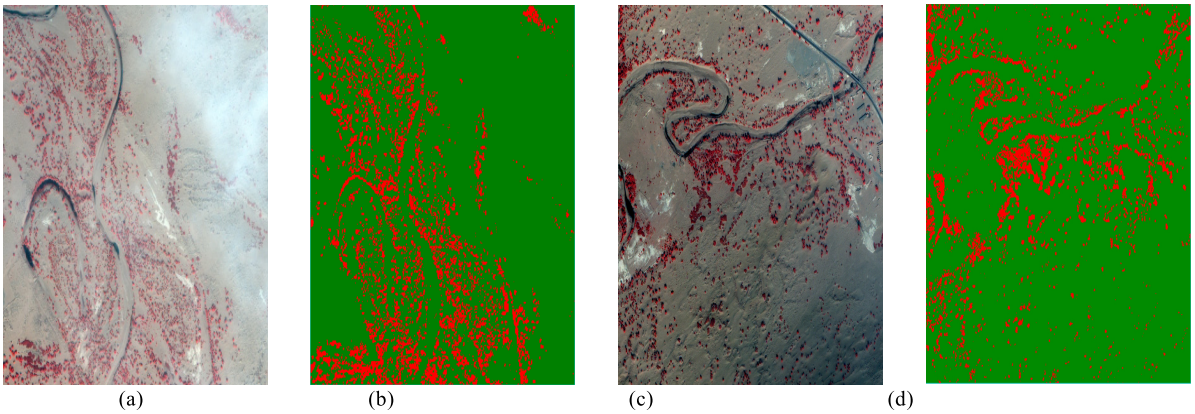


Fig. 5. Quick Bird satellite set. (a) Original image of Luntai County (data set 1). (b) Ground truth of Luntai County (data set 1). (c) Original image of Tarim River Basin (data set 2). (d) Ground truth of Tarim River Basin (data set 2).

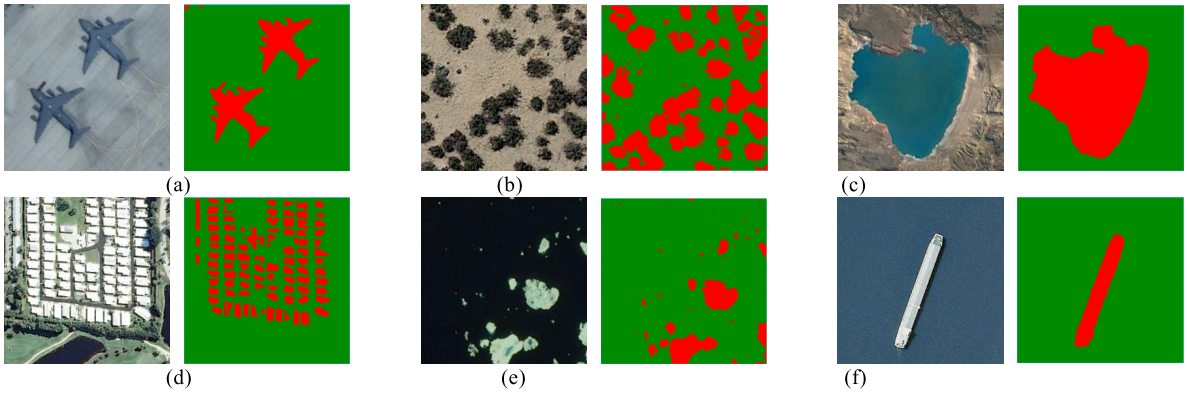


Fig. 6. NWPU-RESISC45 set, including (Left) original images and (Right) their ground truth. (a) Airplane_150. (b) Chaparral_041. (c) Lake_006. (d) Mobile_home_park_190. (e) Sea_ice_304. (f) Ship_048.

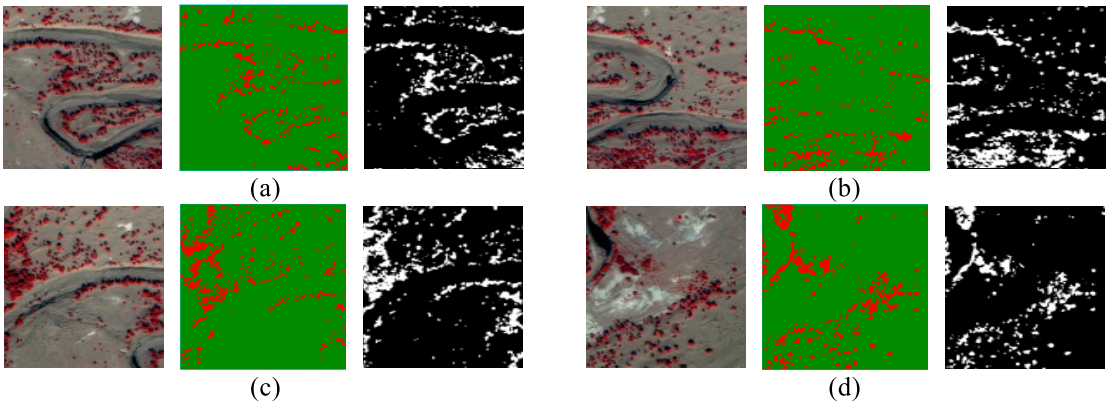


Fig. 7. Partial visualization of images. (a)–(d) All different parts of original image for Luntai County. (First column) Original images. The red parts correspond to the Populus Euphratica Forest, the black parts to shadows, and the gray parts to land. (Second column) Ground truth of the image. (Third column) Model prediction results. The white areas correspond to the Populus Euphratica Forest and the black to land.

analysis are given in Section IV-G. In Fig. 7, we present partial results of the detection of Populus euphratica forest. To more clearly describe the SAGP model and demonstrate the advantages of the bidirectional word vector feature proposed in this article, we visualize the images with a size of 100 × 100 starting from Section IV-D.

B. Hyperparameter Settings

The values of the hyperparameters affect not only the description of the model but also the use and mining of image detail information. The parameters are set as listed in Table I.

The Adam optimizer is used in this model, which is simple and efficient and is not affected by gradient expansion during parameter updating. The parameters are as follows: LR denotes the learning rate; Beta-1 denotes the exponential decay rate of the first moment estimate; Beta-2 denotes the second-order exponential decay rate of the moment estimate; Epsilon is set to 1^{e-08} instead of 0 to prevent the denominator from being 0; and Decay represents the learning rate attenuation value after each parameter update. The parameters of the two-layer BiIndRNNs (SRNN-1 and SRNN-2) are as follows: units represent the number of neurons, and recurrent-clip-min

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TABLE I
HYPERPARAMETER SETTINGS

Type	Hyperparameter
Adam	LR = 0.001 Beta-1 = 0.9 Beta-2 = 0.999 Epsilon = 1^{e-08} Decay = 3^{e-8}
SRNN-1 (BiIndRNN)	units=54 recurrent-clip-min=-1 recurrent-clip-max=-1 dropout=0.35 recurrent-sequence=True
SRNN-2 (BiIndRNN)	units=54 recurrent-clip-min=-1 recurrent-clip-max=-1 dropout=0.35 recurrent-sequence=True
SRNN-3 (Conv)	filters = 128/128/64/64 kernel-size = 5/5/3/3 strides = 1/1/1/1 padding=valid/valid/valid/valid
SRNN-3 (Maxpooling)	pool-size =2/2 padding=valid/valid
Attention	Bias = True
GraphCov	filters=128/128 neighbors-ix-mat=8/4 num-neighbors = 25/16
Dense layer	Units =64/2

and recurrent-clip-max indicate the use of the default time step. The bias term in the attention algorithm is set to 0. The GCN module uses a two-layer graph convolution with the following parameters: neighbors-ix-mat indicates that the current feature is associated with the surrounding n^2 features, and num-neighbors represents the retention of m features from the n^2 feature and will be used as the input to the next layer of GCN.

C. Experimental Evaluation Criteria

This experiment used a variety of evaluation indicators including the precision (P), recall (R), and F-measure (F) values and Accuracy to verify the experiment. P represents the actual proportion of positive examples in the examples that are divided into positive examples; R is a measure of coverage, and multiple positive examples of metrics are divided into positive examples; F is an evaluation index that comprehensively considers the positive and negative samples, which can better reflect the model detection effect; the indicator Accuracy refers to the accuracy, considering the positive and negative probabilities of the model. TN represents the correct number of negative sample classification, FN represents the incorrect number of negative sample classification, FP represents the incorrect number of positive sample classification, and TP represents the correct number of positive sample classification. The calculation formulas are shown as follows:

$$P = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{16}$$

$$R = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{17}$$

$$F = (2P * R)/(P + R) \tag{18}$$

$$\mathrm{Accuracy} = (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN}). \tag{19}$$

D. Impact of the Data Features

The features that are extracted in this article are the color features (HSV), texture features (T), and spatial relationship features (S). The spatial relationship features are mapped into a word vector in Glove and Word2Vec. The naming convention of the feature combinations is as follows: the Glove horizontal word vector is denoted by S-G-Hor; the Glove vertical word vector is denoted by S-G-Ver; the Glove bidirectional word vector is denoted by S-G-All; the Word2Vec horizontal word vector is denoted by S-W-Hor; the Word2Vec vertical word vector is denoted by S-W-Ver; and the Word2Vec bidirectional word vector is denoted by S-W-All). The final features are denoted either All-Feature-G (HSV + T + S-G) in Glove or All-Feature-W (HSV + T + S-W) in Word2Vec. To evaluate the generalization ability of the features in this experiment, three types of segmentation methods are applied to the two data sets. The experimental results are listed in Table II.

Different features and different vector blocks can learn different information. According to Table II, the bold values represent the experimental results of the bidirectional word vector and the red dotted boxes represent the experimental results of all the features. The word vector of the spatial relationship feature map outperforms the original feature in mining the potential features among the image pixels. From Table II, we can see that if we only take the single spatial relation feature as the input, the P value is relatively low. After the original spatial relation feature is generated by the Word2vec and Glove algorithms, the P value greatly improves. The reason for low P value indicates that the result of learning and classifying samples only through spatial relationship features or texture features is poor. The real-valued vector that is mapped by the spatial relationship feature word is improved by 1%–1.5% on average compared with the original spatial relationship feature. The bidirectional word vector in Glove slightly outperforms the Word2Vec representation, which demonstrates the importance of global features. The bidirectional word vector method outperforms the traditional one-way word vector method, and the accuracy rate is improved by approximately 1%–3%. Therefore, bidirectional word vectors can be used to identify potential features between image pixels. By combining various features, this article demonstrates that multiple features outperform single features. In both data sets, the accuracy rate is 0.5%–1% higher than that from the best single feature. When an image is cut into 3 × 3 vectorized image blocks, it is difficult to describe the image by texture features only, but the spatial relationship feature can learn the deep features to improve the accuracy even for a small block size. When an image is cut into 5 × 5 vectorized image blocks, certain potential features of the image can be learned through the texture features, with accuracies of 79.01% and 88.61% for data set 1 and data set 2, respectively. The spatial relationship features also learn more potential features with increasing block size, with accuracies of 82.13% and 93.38%, respectively. When the image is cut into 7 × 7 vectorized image blocks, the texture features can learn additional potential features of

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TABLE II
EXPERIMENTAL RESULTS FOR VARIOUS FEATURES

Features	Image block size	Dataset 1				Dataset 2			
		Accuracy	P	R	F	Accuracy	P	R	F
HSV	3*3	0.9316	0.8244	0.8053	0.8148	0.9433	0.874	0.7873	0.8286
T		—	—	—	—	—	—	—	—
S		0.784	0.4429	0.7166	0.5474	0.8776	0.44	0.9199	0.5953
S-W-Hor		0.8193	0.5081	0.7069	0.5912	0.882	0.4535	0.7836	0.5745
S-W-Ver		0.8036	0.468	0.7094	0.564	0.897	0.4821	0.7372	0.5829
S-W-All		0.8336	0.5594	0.6293	0.5923	0.9136	0.5444	0.7474	**0.63**
S-G-Hor		0.8241	0.513	0.6648	0.5791	0.9018	0.4924	0.7947	0.6081
S-G-Ver		0.8268	0.5099	0.651	0.572	0.893	0.4768	0.8699	0.616
S-G-All		**0.8476**	0.5635	0.6293	**0.5947**	**0.92**	0.5714	0.6896	0.625
All-Feature-W		0.9445	0.8848	0.8062	0.8437	0.9671	0.8476	0.7918	0.8187
All-Feature-G		0.945	0.8883	0.8076	0.8461	0.9677	0.8469	0.8212	0.8339
HSV	5*5	0.943	0.7331	0.9213	0.8165	0.9641	0.7479	0.8846	0.8105
T		0.7901	0.3101	0.4687	0.3733	0.8861	0.3387	0.2707	0.3009
S		0.8213	0.4594	0.7571	0.5715	0.9338	0.5324	0.7334	0.617
S-W-Hor		0.8255	0.4088	0.7407	0.5268	0.938	0.595	0.7107	0.6481
S-W-Ver		0.8396	0.4181	0.5836	0.4872	0.9373	0.6103	0.686	0.6459
S-W-All		0.8563	0.5039	0.5562	0.5287	0.9408	0.6156	0.7372	0.6709
S-G-Hor		0.8511	0.4864	0.6205	0.5453	0.942	0.6243	0.7664	0.6881
S-G-Ver		0.853	0.4642	0.6388	0.537	0.9433	0.602	0.6763	0.637
S-G-All		**0.8613**	0.4872	0.6385	**0.5526**	**0.9493**	0.6884	0.7024	**0.6953**
All-Feature-W		0.959	0.846	0.8543	0.8502	0.9723	0.8235	0.827	0.8252
All-Feature-G		0.966	0.8806	0.8458	0.8629	0.9728	0.8261	0.8718	0.8483
HSV	7*7	0.962	0.8224	0.9175	0.8674	0.9703	0.7303	0.8918	0.8031
T		0.9505	0.7384	0.9603	0.8349	0.9611	0.6494	0.9441	0.7695
S		0.8668	0.4848	0.733	0.5836	0.944	0.541	0.7408	0.6253
S-W-Hor		0.8805	0.5367	0.6318	0.5804	0.954	0.6551	0.6902	0.6722
S-W-Ver		0.8798	0.5246	0.651	0.5811	0.9508	0.6489	0.7566	0.6986
S-W-All		0.886	0.5461	0.6204	0.5937	0.9576	0.7512	0.6548	0.6997
S-G-Hor		0.8843	0.5367	0.6318	0.5641	0.957	0.6967	0.6698	0.683
S-G-Ver		0.8838	0.5345	0.528	0.5312	0.9541	0.6652	0.7314	0.6968
S-G-All		**0.8867**	0.5433	0.6675	**0.5991**	**0.958**	0.743	0.6879	**0.7143**
All-Feature-W		0.9675	0.8764	0.8586	0.8674	0.978	0.8197	0.8817	0.8496
All-Feature-G		0.9712	0.8871	0.8631	0.8749	0.9808	0.8771	0.8459	0.8612

the image. The accuracies increase to 95.05% and 96.11%, respectively; however, no significant difference is seen for the spatial relationship features, and the accuracies are 88.6% and 94.4%, respectively. Hence, the bidirectional word vector fully exploits the spatial relationship features of the image.

The results in Table II indicate that combining the bidirectional word vectors with the original image features helps to retrieve the potential features of the images by integrating contextual information as much as possible. As illustrated in Fig. 8, the diagrams display the feature effect by the word vectors in various image block sizes (the numbers in front of the feature denotation specify the size of the image block size).

As shown in Fig. 8, when 3×3 vectorized image blocks are used to generate the word vector maps, more noise is generated. In addition, the structure diagram that is generated by Word2Vec has more noise than that by Glove, which fully demonstrates the importance of global features. As the size of the vector image block increases, the noise gradually decreases, and the resulting image is more compact and coherent.

E. Impact of Various Combination of the Model

Models differ in their effects on data mining, and the joint model outperforms the single model. We use the following naming convention in specifying the models: SRNN represents a single SRNN; A-GCN represents the attention mechanism treated GCNs; SRNN-GCN(1) represents the sequential joint extraction via SRNN and the GCN algorithm; SRNN-GCNs(2) represents the parallel joint extraction via SRNN and the GCN algorithm; and SRNN-A-GCNs(1) represents the sequential joint extraction via SRNN, the attention mechanism, and the GCN algorithm. The results are listed in Table III (using 7×7 block size with the bidirectional Glove feature and the traditional feature).

According to Table III, the sequential joint model outperforms the single-structure neural network model and can fully capture the deep features of the image; the accuracies reach from 96.25% to 97.12% for data set 1 and from 96.76% to 98.08% for data set 2. The proposed SAGP model has the highest accuracy, and P and F values for both data set 1 and data set 2, which demonstrates that the parallel joint model can better integrate the potential features of the image. The semantic information of the context yields more accurate results: the accuracies are 97.12% and 98.08% for data set 1 and data set 2, respectively, which is 0.87% and 1.27% higher than those of the traditional combination model, respectively. We compare with the state-of-the-art methods, including SRNN, GCN, and attention; the results show that

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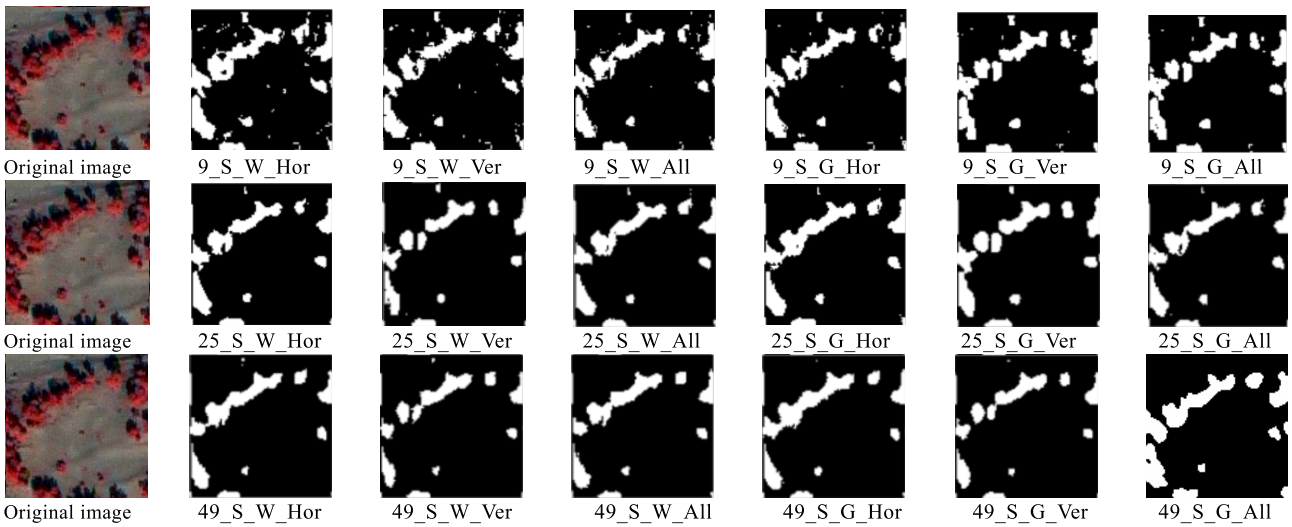


Fig. 8. Vector image blocks generated by various word vectors. The ground truth is supplemented in Fig. 9.

TABLE III
EXPERIMENTAL RESULTS OF SIX MODEL STRUCTURES

Model	Dataset 1				Dataset 2			
	Accuracy	P	R	F	Accuracy	P	R	F
SRNN[9]	0.8972	0.5916	0.5687	0.5799	0.9506	0.614	0.6572	0.6349
GCN[20]	0.946	0.7195	0.984	0.8312	0.9712	0.8662	0.8509	0.8585
Attention[42]	0.8643	0.4987	0.809	0.617	0.9139	0.5661	0.7094	0.6297
A-GCNs	0.9563	0.8523	0.8068	0.8289	0.9698	0.7311	0.8374	0.7807
SRNN-GCNs(1)	0.9625	0.8084	0.9189	0.8601	0.9676	0.7042	0.9588	0.8120
SRNN-GCNs(2)	0.9685	0.7454	0.9581	0.838	0.9778	0.8364	0.8503	0.8433
SRNN-A-GCNs(1)	0.9441	0.7249	0.9103	0.8071	0.9681	0.7299	0.9029	0.8073
SAGP	0.9712	0.8871	0.8631	0.8749	0.9808	0.8771	0.8459	0.8612

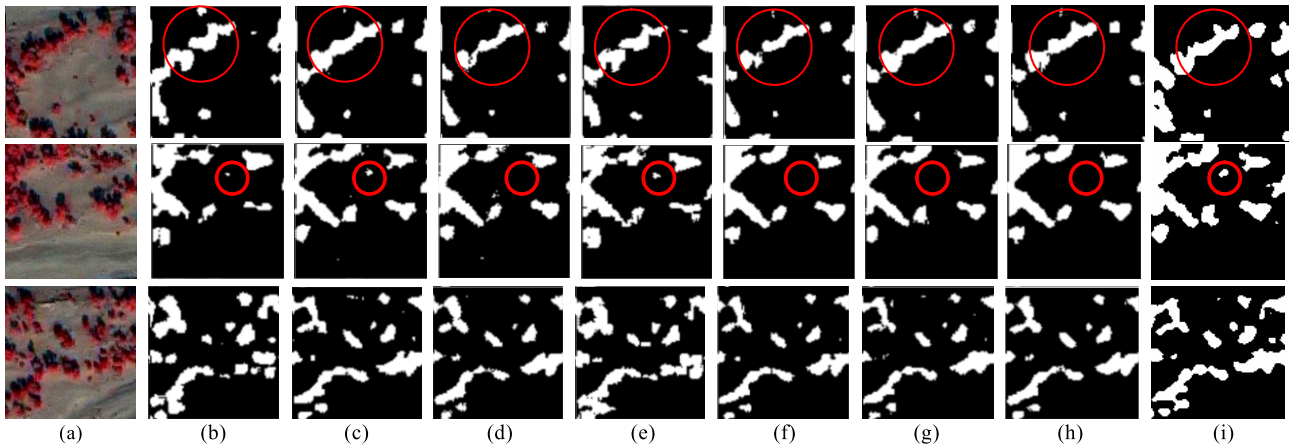


Fig. 9. Renderings of various models. The ground truth is supplemented. (a) Original image. (b) SRNN. (c) GCN. (d) Attention. (e) A-GCNs. (f) SRNN-GCNs(1). (g) SRNN-GCNs(2). (h) SRNN-A-GCNs(1). (i) SAGP.

SAGP is optimal. In Fig. 9, the diagrams illustrate the feature effects generated by various models (substantial differences are indicated by red circles).

According to the first set of images in Fig. 9, as the structure of the model becomes more complex, the deep features of the image can be better retrieved, thereby resulting in a clearer outline and superior information integration. According to the second set of images, the proposed model exhibits superior performance in the mining of small targets and is more suitable for capturing the global deep features of the image. The images in the third group are more complicated; the SAGP method more accurately describes the content of the images, and the effect on noise reduction is substantial. The SAGP model outperforms the other models, which demonstrates that the SAGP model can more accurately capture the deep features.

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TABLE IV
EXPERIMENTAL RESULTS AT VARIOUS LAYERS

Model	Dataset 1				Dataset 2			
	Accuracy	P	R	F	Accuracy	P	R	F
GCNs 1 layer	0.962	0.8265	0.8834	0.8540	0.9743	0.8596	0.7729	0.814
GCNs 3 layers	0.9595	0.8233	0.8789	0.8502	0.975	0.7874	0.9049	0.8421
BiIndRNN 1 layer	0.9578	0.7737	0.92	0.8405	0.9768	0.8599	0.8146	0.8366
BiIndRNN 3 layer	0.9513	0.7363	0.9372	0.8247	0.9773	0.7955	0.8872	0.8388
SAGP	0.9712	0.8871	0.8631	0.8749	0.9808	0.8771	0.8459	0.8612

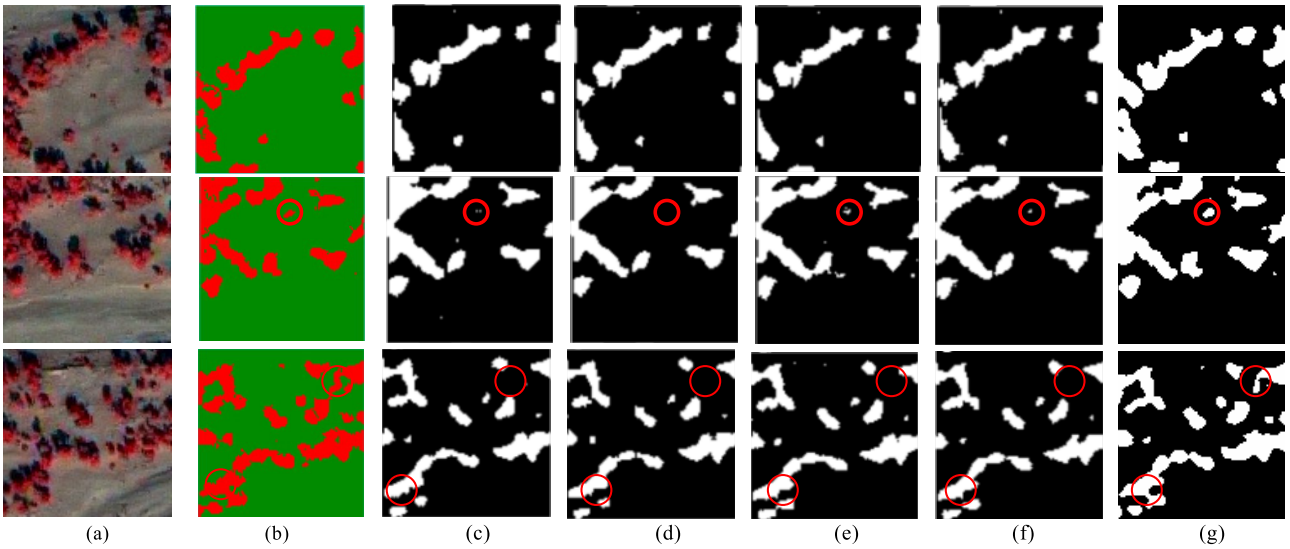


Fig. 10. Effect diagram of various model layer generation approaches. (a) Original image. (b) Ground truth. (c) GCNs 1 layer. (d) GCNs 3 layers. (e) BiIndRNN-1. (f) BiIndRNN-3. (g) SAGP.

F. Influence of the Number of Layers in the Model

The number of model layers has various effects on the experimental data. In this article, the best-performing data are used to evaluate the model selection, and the experimental data are mined by increasing or decreasing the number of GCNs and the number of BiIndRNN columns that are juxtaposed under SRNN to understand the impact. "BiIndRNN 3 layer" indicates that the third column of SRNN uses BiIndRNN. The experimental results are listed in Table IV.

From Table IV, the deep features that are extracted by the CNN and the context relationships that are established by the RNN can jointly mine the intrinsic information of the image better. When the GCNs have two layers, and two columns of BiIndRNN and one column CNN are juxtaposed in the SAGP model, the best results are obtained. The experimental results demonstrate that when the number of GCN layers increases, the model loses deep features, and the accuracies are 95.95% and 97.5% for data sets 1 and 2, respectively; when the number of GCNs is reduced, the model algorithm cannot fully exploit the deep features and describe the image details, and the accuracies are 96.2% and 97.43% for data sets 1 and 2, respectively. Replacing the last layer of the BiIndRNN with the CNN yields more accurate results: the accuracies are 97.12% and 98.08%, respectively. Hence, the deep features that are extracted by the CNN and the context relationships that are established by the RNN can jointly retrieve the intrinsic information of the image better. As illustrated in Fig. 10, the diagrams display the feature effects when various model layers are used. (The substantial differences are indicated by red circles.)

According to Fig. 10, when the number of model layers is too low, additional noise is generated, and because of the noise, these features are very poor for describing the image. In contrast, when the number of model layers is too high, deep features are lost; the excessive loss of deep features also makes these features very poor for describing the image. The defects in the second set of images are more easily observed, and the results generated are of poor quality. In Fig. 10, the saturation of the real target is relatively poor. When the third layer of the SRNN is changed to a CNN, the CNN retrieves the deep features of the original image in conjunction with the contextual semantic information from the BiIndRNN network. Both the deep feature mining and the resulting image saturation are superior compared with the traditional SRNN algorithm.

G. Effect of Window Size

The size of the window directly reflects the accuracy of the relationship between the current pixel and the surrounding pixels. If the window is too large, the deep pixel weight weakens, and the complexity of the training increases. If the window is too small, the direct relationship of the pixels is not correctly reflected. The experimental results are listed in Table V.

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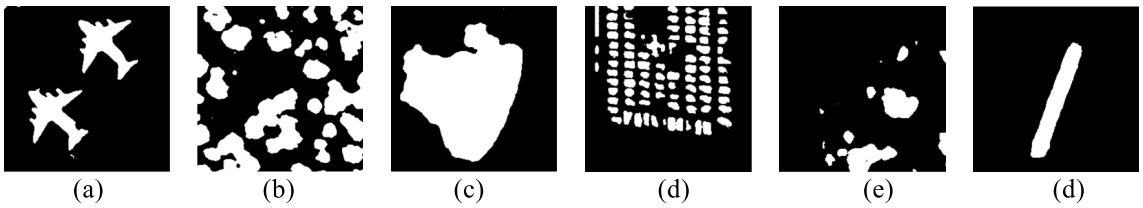


Fig. 11. Effect diagrams of NWPU-RESISC45 data set. (a) Airplane_150. (b) Chaparral_041. (c) Lake_006. (d) Mobile_home_park_190. (e) Sea_ice_304. (f) Ship_048.

TABLE V
EXPERIMENTAL RESULTS OF FIVE WINDOW SIZES

Window Size	Dataset 1			
	Accuracy	P	R	F
Window-size = 3	0.9598	0.7935	0.8679	0.8291
Window-size = 4	0.9637	0.7976	0.9231	0.8558
Window-size = 5	0.9712	0.8871	0.8631	0.8749
Window-size = 6	0.9718	0.9178	0.8225	0.8676
Window-size = 7	0.9556	0.789	0.8256	0.8069
Window Size	Dataset 2			
	Accuracy	P	R	F
Window-size = 3	0.9603	0.8408	0.7984	0.819
Window-size = 4	0.9711	0.8562	0.8874	0.8715
Window-size = 5	0.9808	0.8771	0.8459	0.8612
Window-size = 6	0.9739	0.9299	0.8296	0.8769
Window-size = 7	0.9618	0.8726	0.7731	0.8199

TABLE VI
EXPERIMENTAL RESULTS FROM NWPU-RESISC45

Model	NWPU-RESISC45			
	Accuracy	P	R	F
airplane_150	0.9891	0.9911	0.9226	0.9556
chaparral_041	0.9816	0.9579	0.9907	0.974
lake_006	0.9967	0.9954	0.9958	0.9956
home_park_190	0.9864	0.9633	0.971	0.9671
sea_ice_304	0.9957	0.9536	0.9951	0.9739
ship_048	0.9992	0.9918	0.9965	0.9941

As shown in Table V, a window size of 5 or 6 yields the best effects, and the direct relationship of the pixels can be better established. When the window size is reduced, it is difficult to sufficiently establish the relationship between the pixels, which reduces the classification performance. When the window size increases, the weight is dispersed, and the noise increases, and thus, the classification performance is worse.

H. NWPU-RESISC45 Data Set Test Results

To further test the SAGP model for robustness and generalization in image recognition, we select certain images from the NWPU-RESISC45 data set for testing. The results are listed in Table VI.

The effect of the SAGP on the public data set NWPU-RESISC45 is shown in Fig. 11. Fig. 11(a) and (e) shows that the recognition effect of the model can reach 99% in terms of accuracy when the difference between the foreground and background is small. Fig. 11(b) and (d) shows that the model recognition of complex images is also quite good, with accuracies reaching 98.64% and 98.16%, respectively. Fig. 11(c) and (f) shows that the model accuracies are 99.67% and 99.92%, respectively, on simple targets.

V. CONCLUSION

Because the vectors generated by traditional features are too simple to describe complex remote sensing image content, certain existing models, such as single models and the sequential joint model, have difficulty in fully exploiting the deep features of images and describing the semantic contextual relations between features in pixels. Therefore, we propose a bidirectional word vector that fully exploits the contextual semantic relationships between the features in pixels and construct a GCN under the attention mechanism such that the features that have high-probability weights are related to each other. At the same time, the deep features and the semantic context information of the features in pixels are fully captured by the improved SRNN. Finally, the deep features of a variety of models are merged, thereby enhancing the image descriptive representations and increasing the classification accuracy.

Although an image can be described in a more detailed way via natural language processing, understanding this approach in general is not intuitive. As a next step, we will further improve the target detection accuracy by using the multimodal mode, in which the natural language and the image can correspond to each other.

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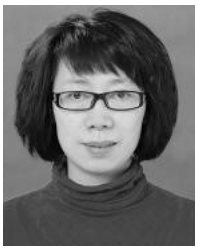
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